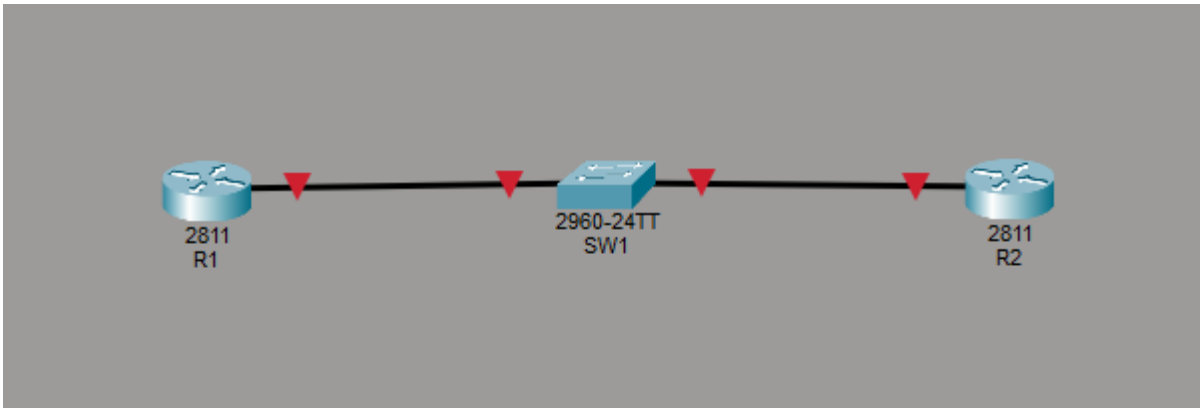
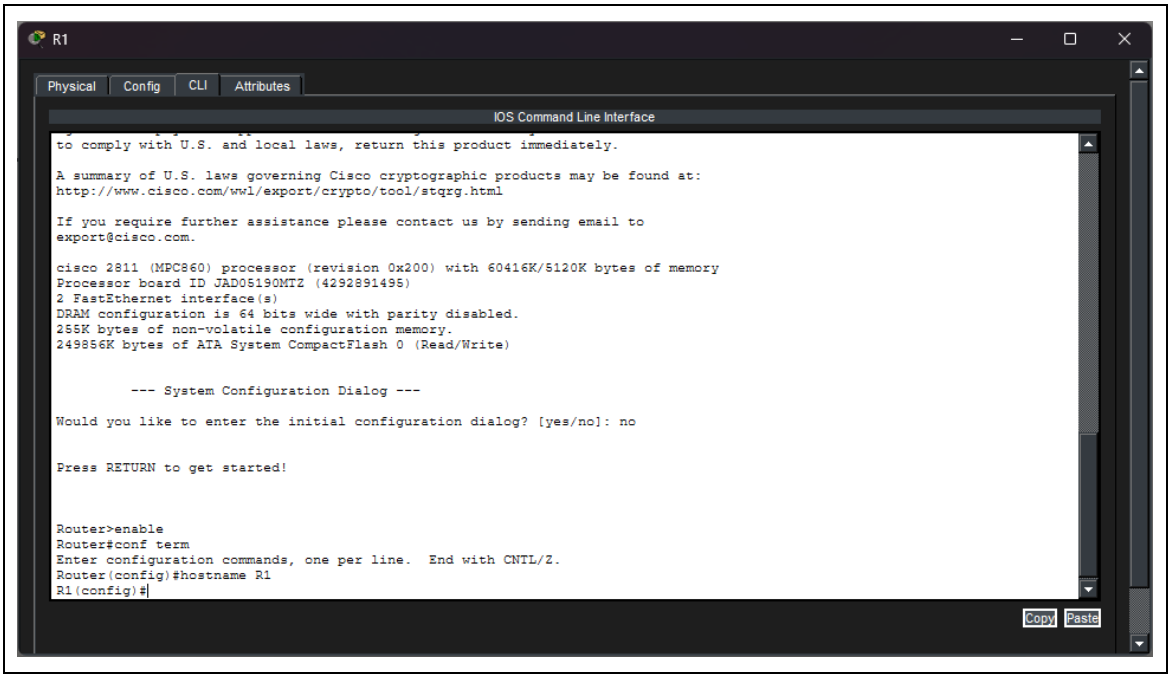
NET512 — Weekly Practical

NAME	Christiaan van Wyk	WEEK	Week 5
STUDENT NO.	20261193	DATE	25 September 2026

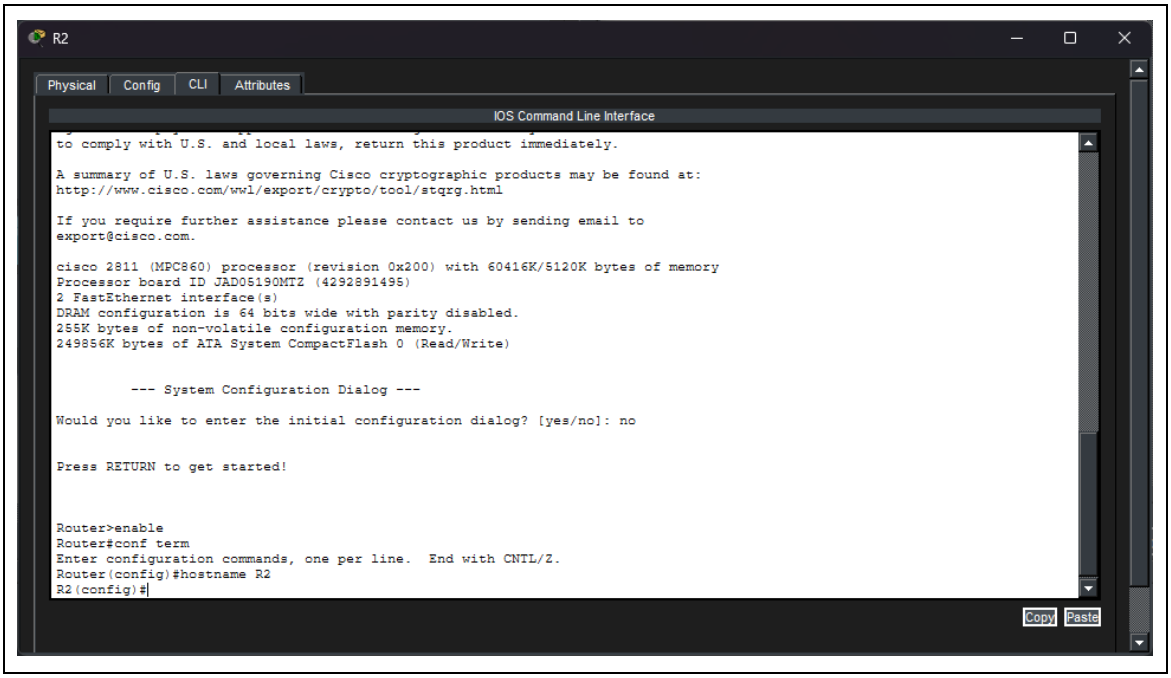
Topology



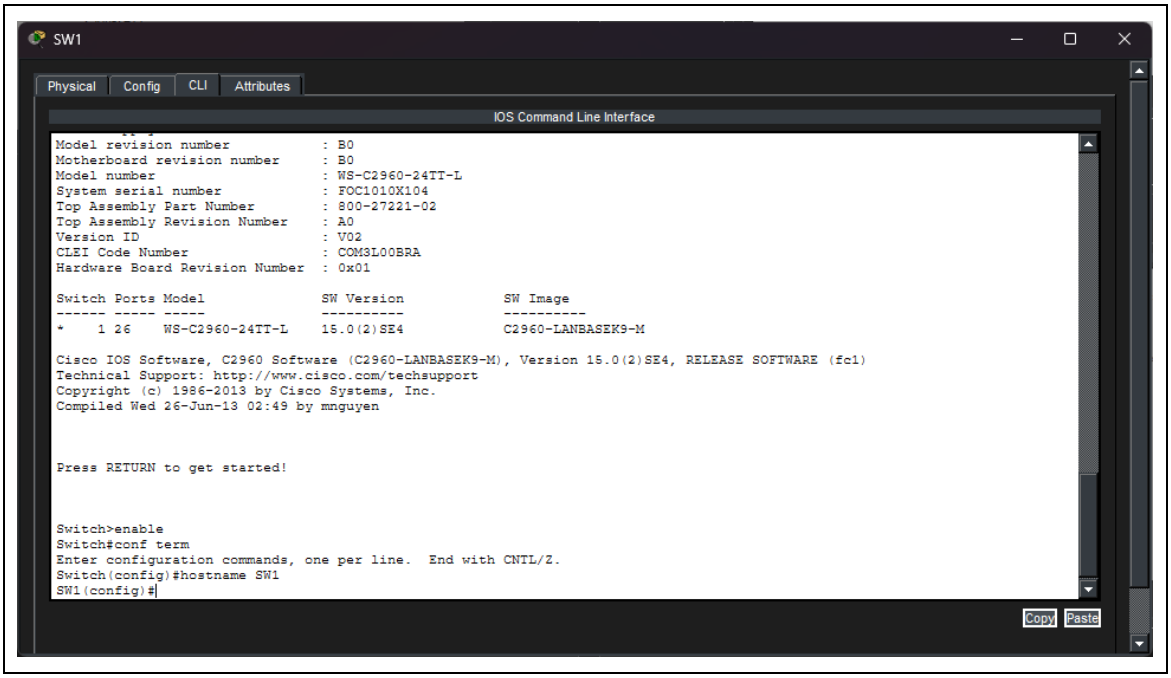
Task 28 — R1 Hostname



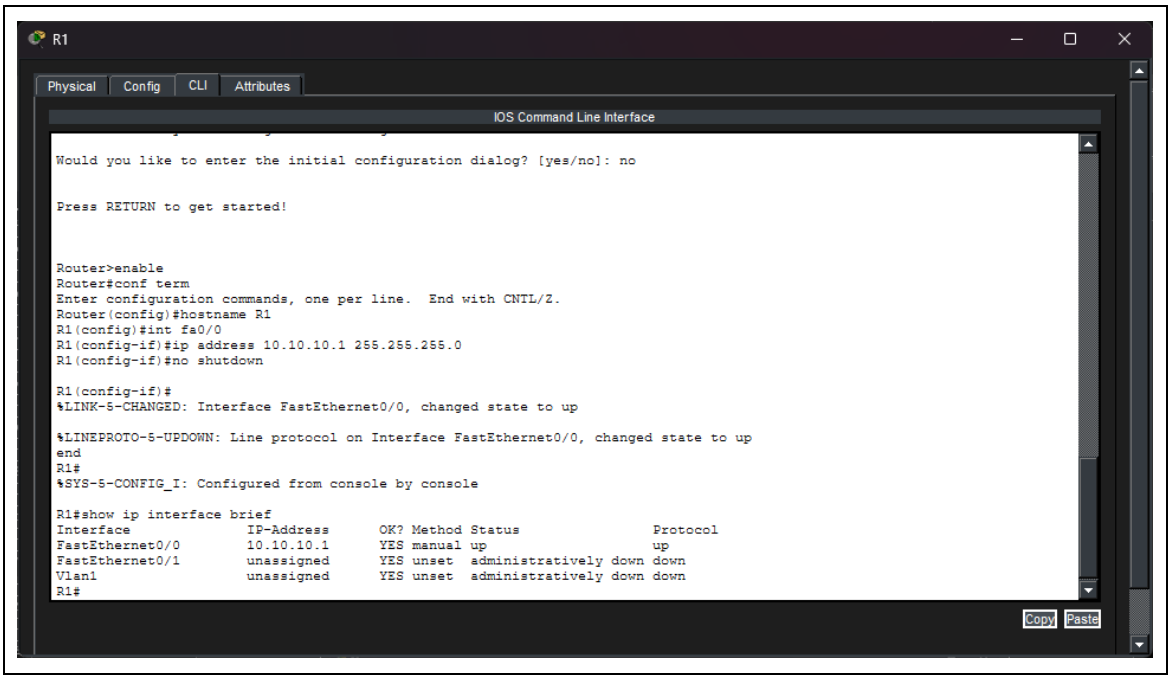
Task 29 — R2 Hostname



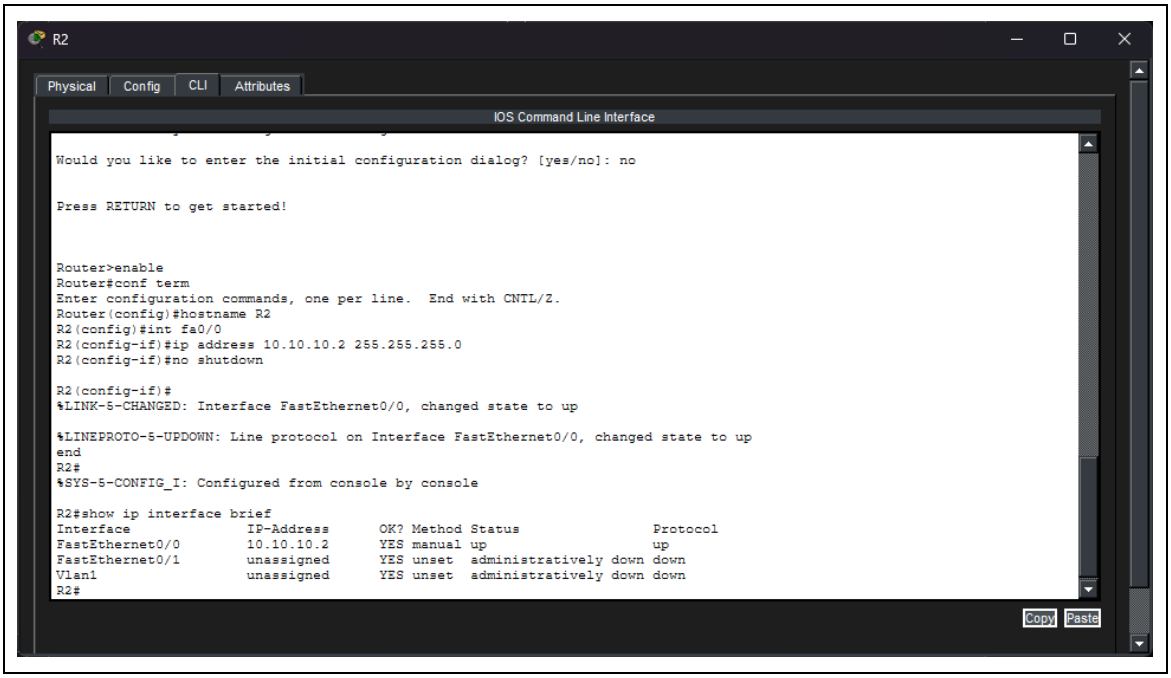
Task 30 — SW1 Hostname



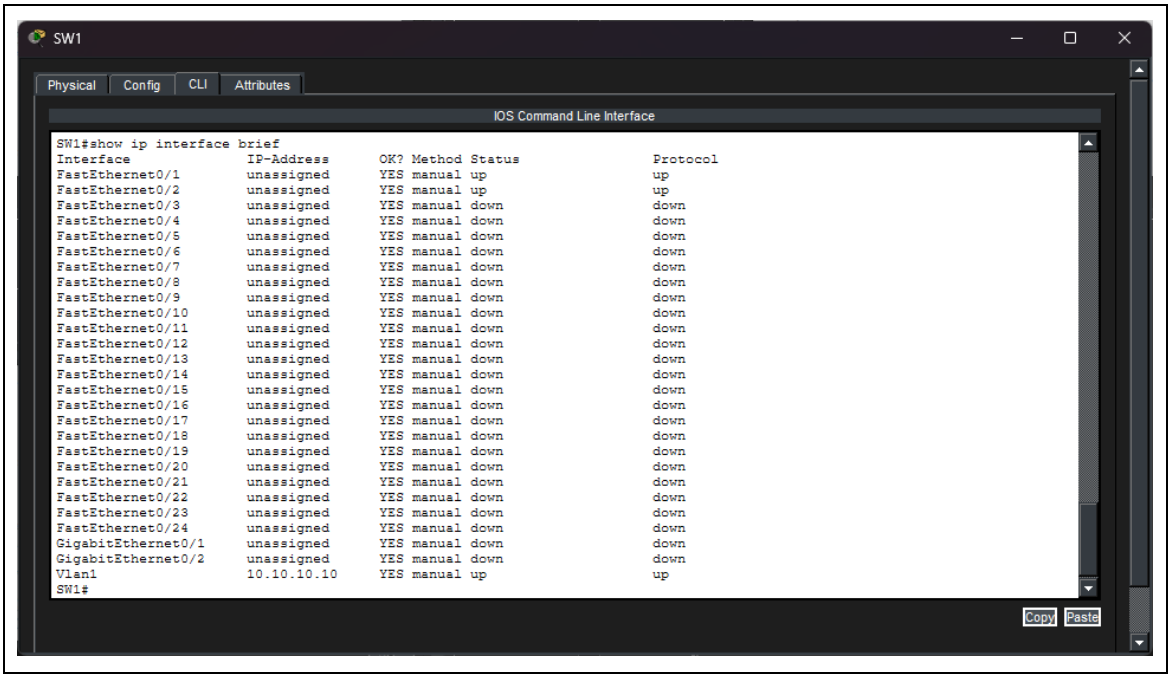
Task 31 — R1 Address



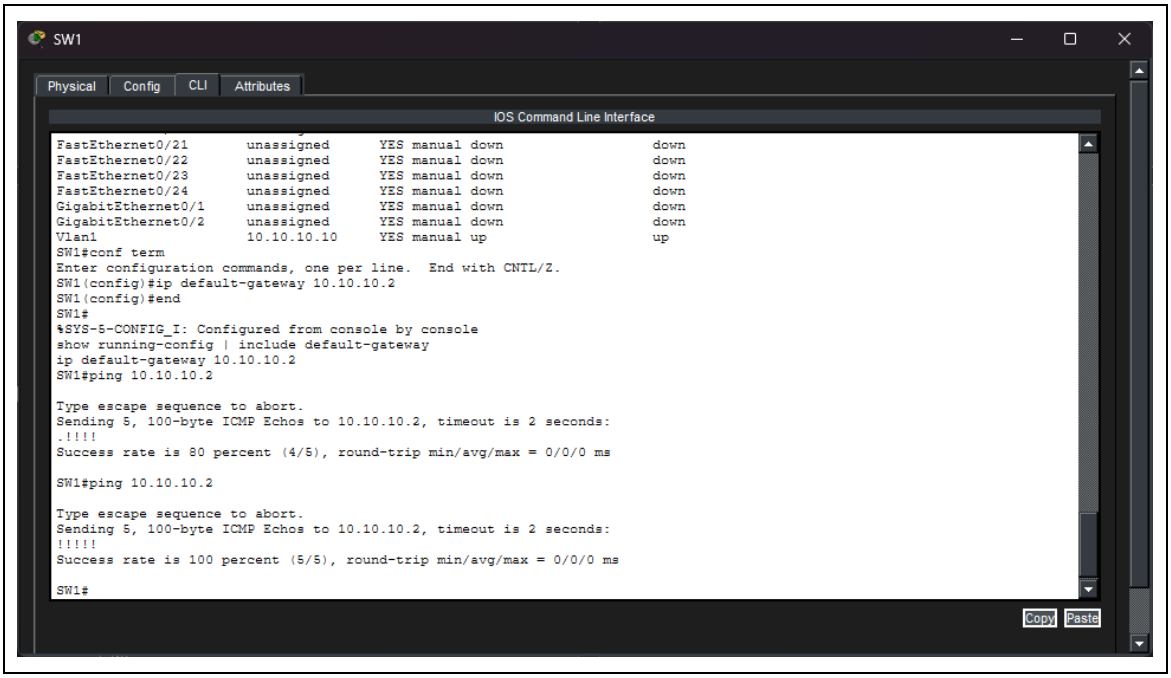
Task 32 — R2 Address



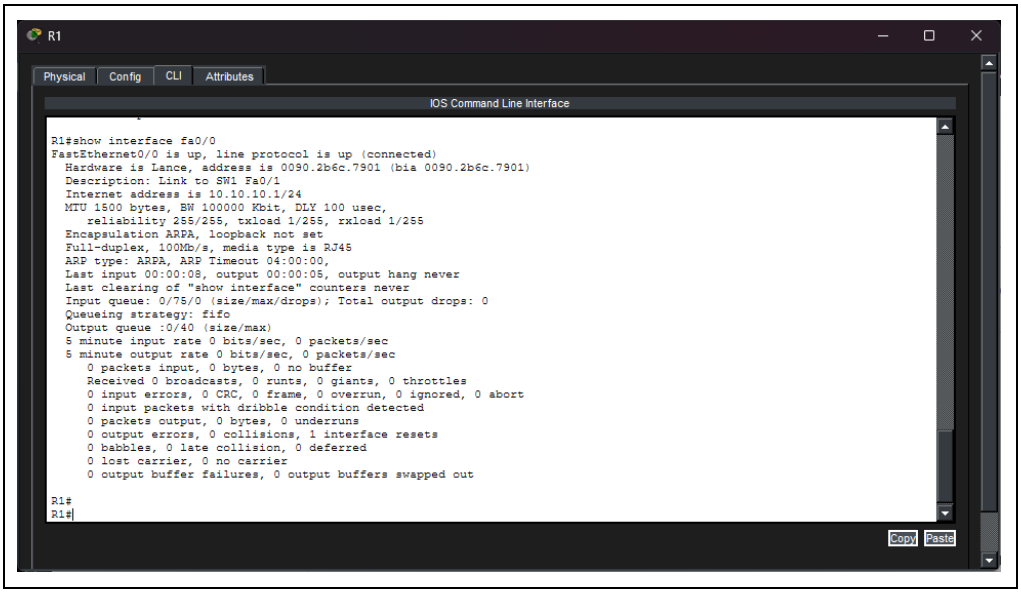
Task 33 — SW1 Management



Task 34 — SW1 Connectivity

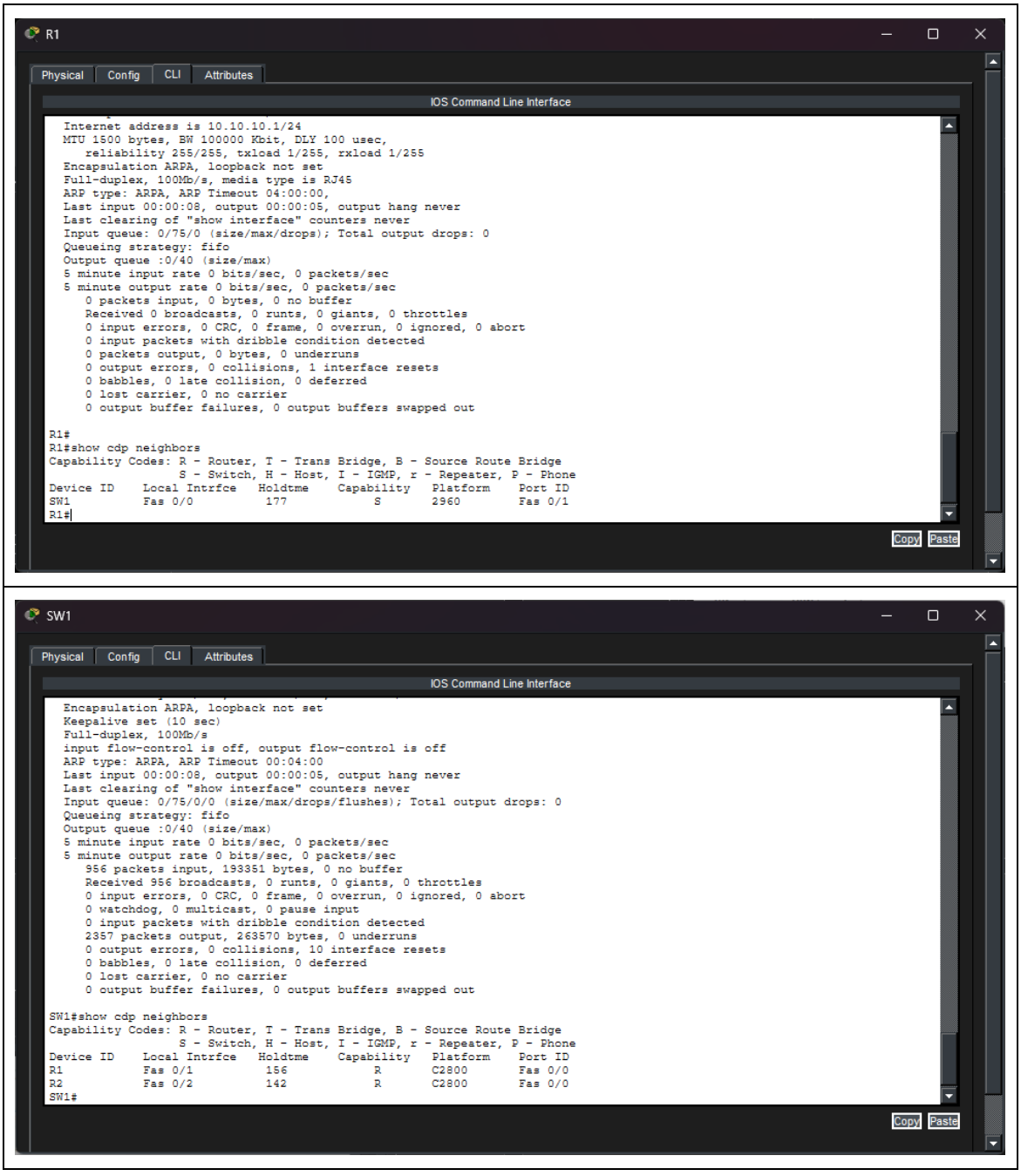


Task 35 — Interface descriptions

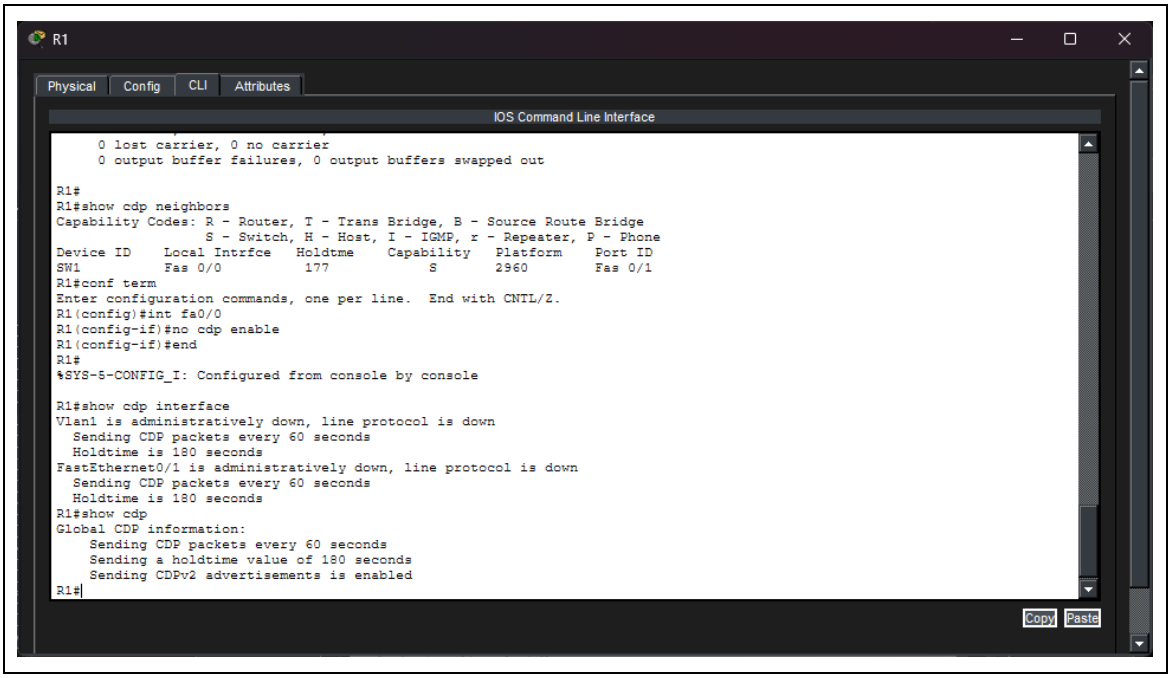



R2
Physical
Config
CLI
Attributes
IOS Command Line Interface
R2#show interface fa0/0
FastEthernet0/0 is up, line protocol is up (connected)
Hardware is Lance, address is 00d0.d366.1001 (bia 00d0.d366.1001)
Description: Link to SW1 Fa0/2
Internet address is 10.10.10.2/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Full-duplex, 100Mb/s, media type is RJ45
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
9 packets input, 1152 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 input packets with dribble condition detected
9 packets output, 1152 bytes, 0 underruns
0 output errors, 0 collisions, 1 interface resets
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
R2#
R2#
Copy
Paste
SW1
Physical
Config
CLI
Attributes
IOS Command Line Interface
SW1#show interface fa0/1
FastEthernet0/1 is up, line protocol is up (connected)
Hardware is Lance, address is 0060.7022.bc01 (bia 0060.7022.bc01)
Description: Link to R1 Fa0/0
BW 100000 Kbit, DLY 100 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s
input flow-control is off, output flow-control is off
ARP type: ARPA, ARP Timeout 00:04:00
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
956 packets input, 193351 bytes, 0 no buffer
Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 watchdog, 0 multicast, 0 pause input
0 input packets with dribble condition detected
2357 packets output, 263570 bytes, 0 underruns
0 output errors, 0 collisions, 10 interface resets
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
SW1#
Copy
Paste
SW1
Physical
Config
CLI
Attributes
IOS Command Line Interface
SW1#show interface fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
Hardware is Lance, address is 0060.7022.bc02 (bia 0060.7022.bc02)
Description: Link to R2 Fa0/0
BW 100000 Kbit, DLY 100 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s
input flow-control is off, output flow-control is off
ARP type: ARPA, ARP Timeout 00:04:00
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
956 packets input, 193351 bytes, 0 no buffer
Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 watchdog, 0 multicast, 0 pause input
0 input packets with dribble condition detected
2357 packets output, 263570 bytes, 0 underruns
0 output errors, 0 collisions, 10 interface resets
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
SW1#
Copy
Paste

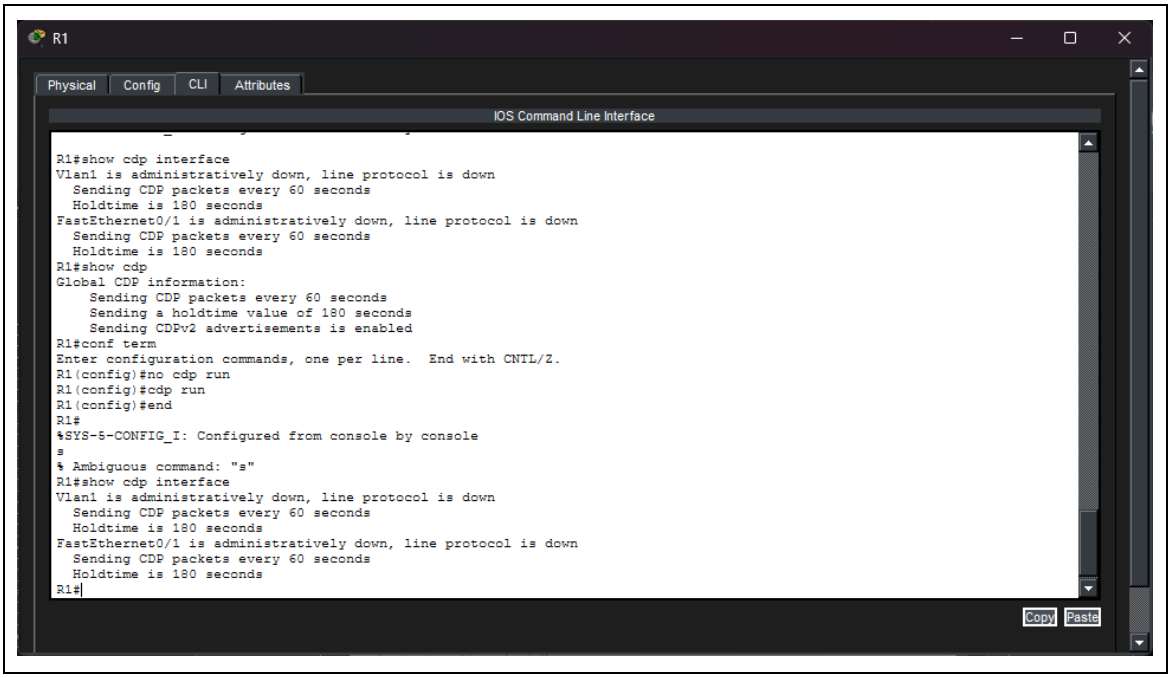
Task 36 — Verify Neighbours



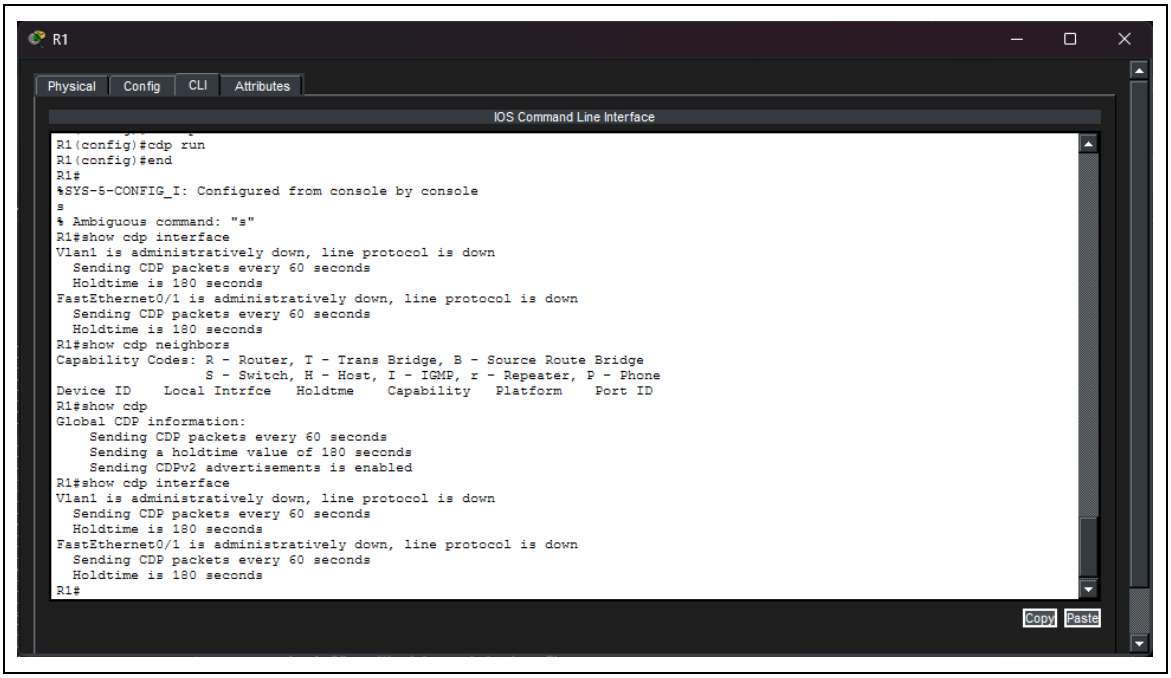
Task 37 — Prevent Discovery



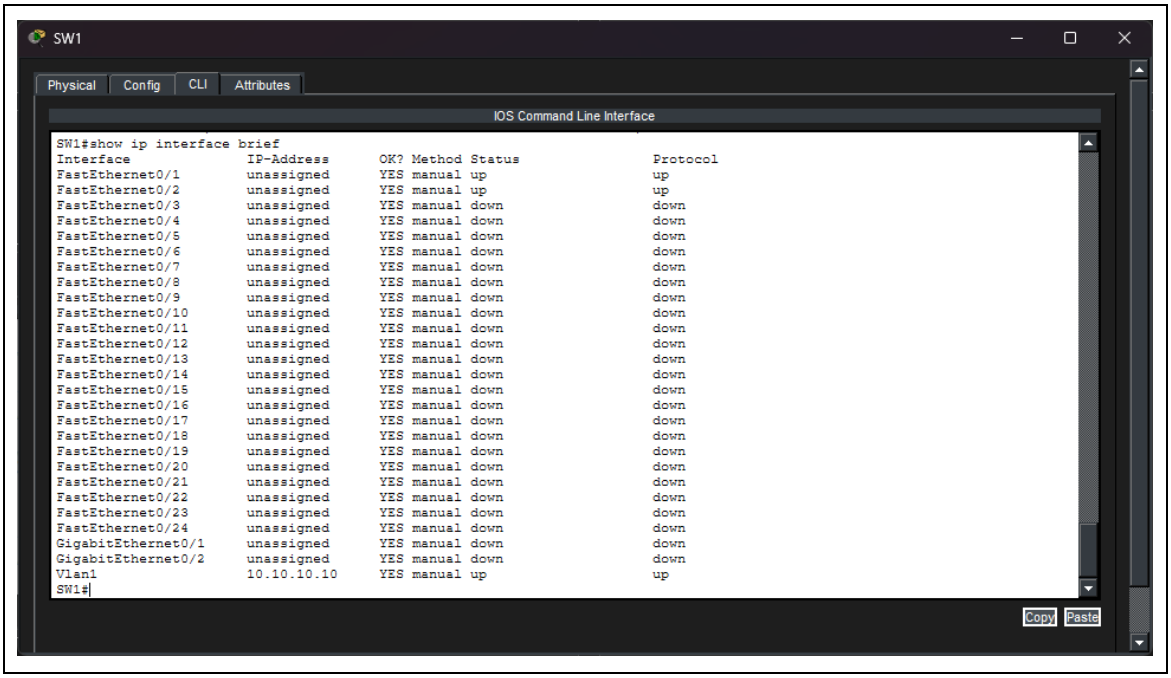
Task 38 — Flush CDP cache



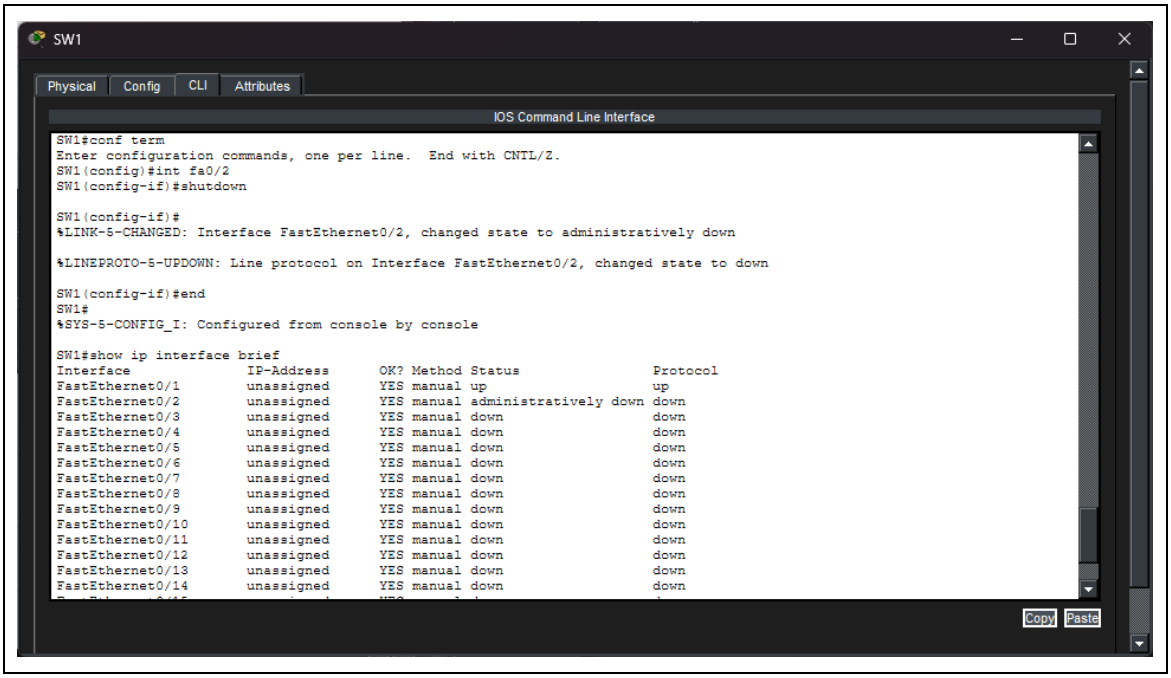
Task 39 — Verify (Non)-Visibility



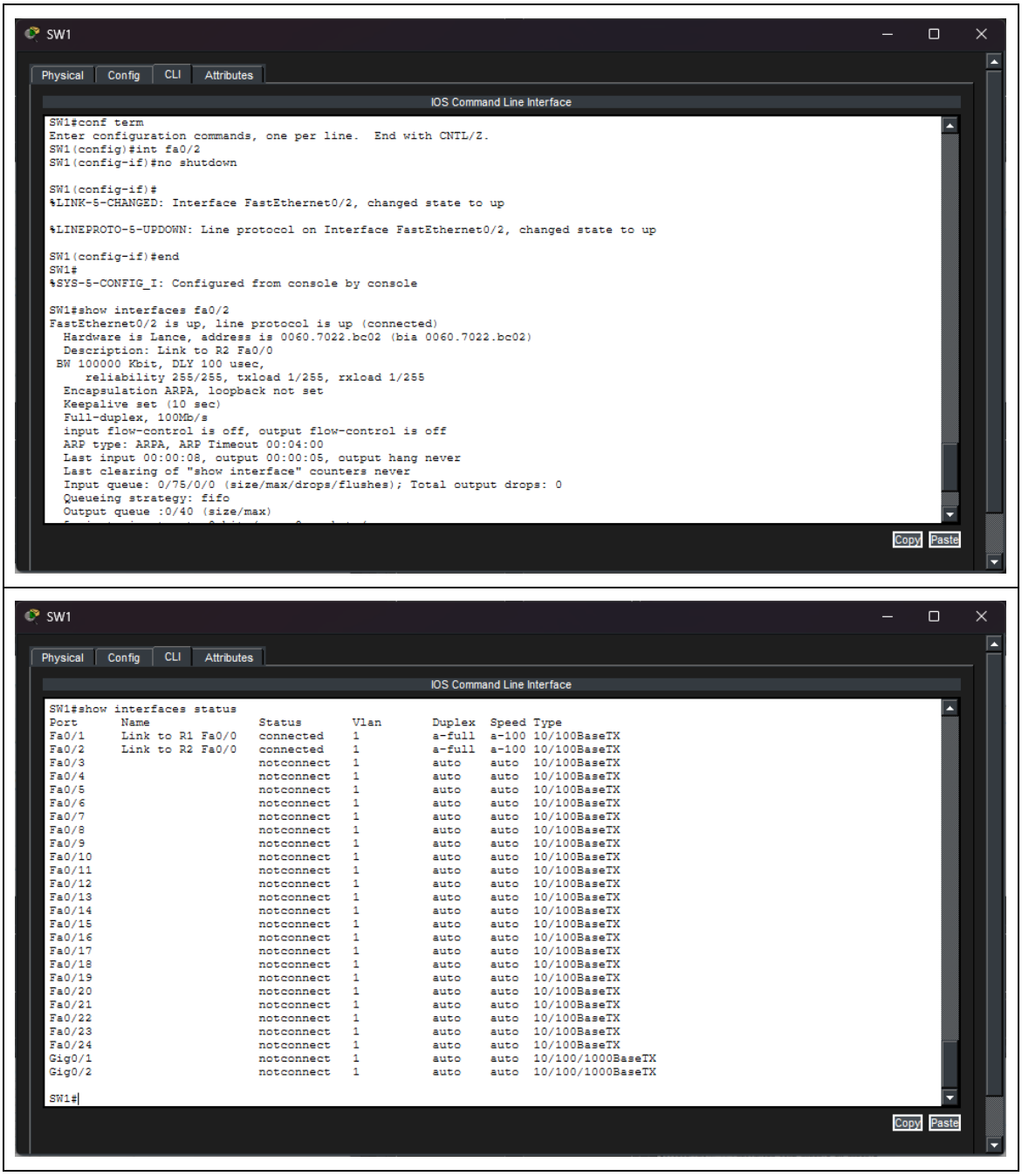
Task 40 — Verify Port Status



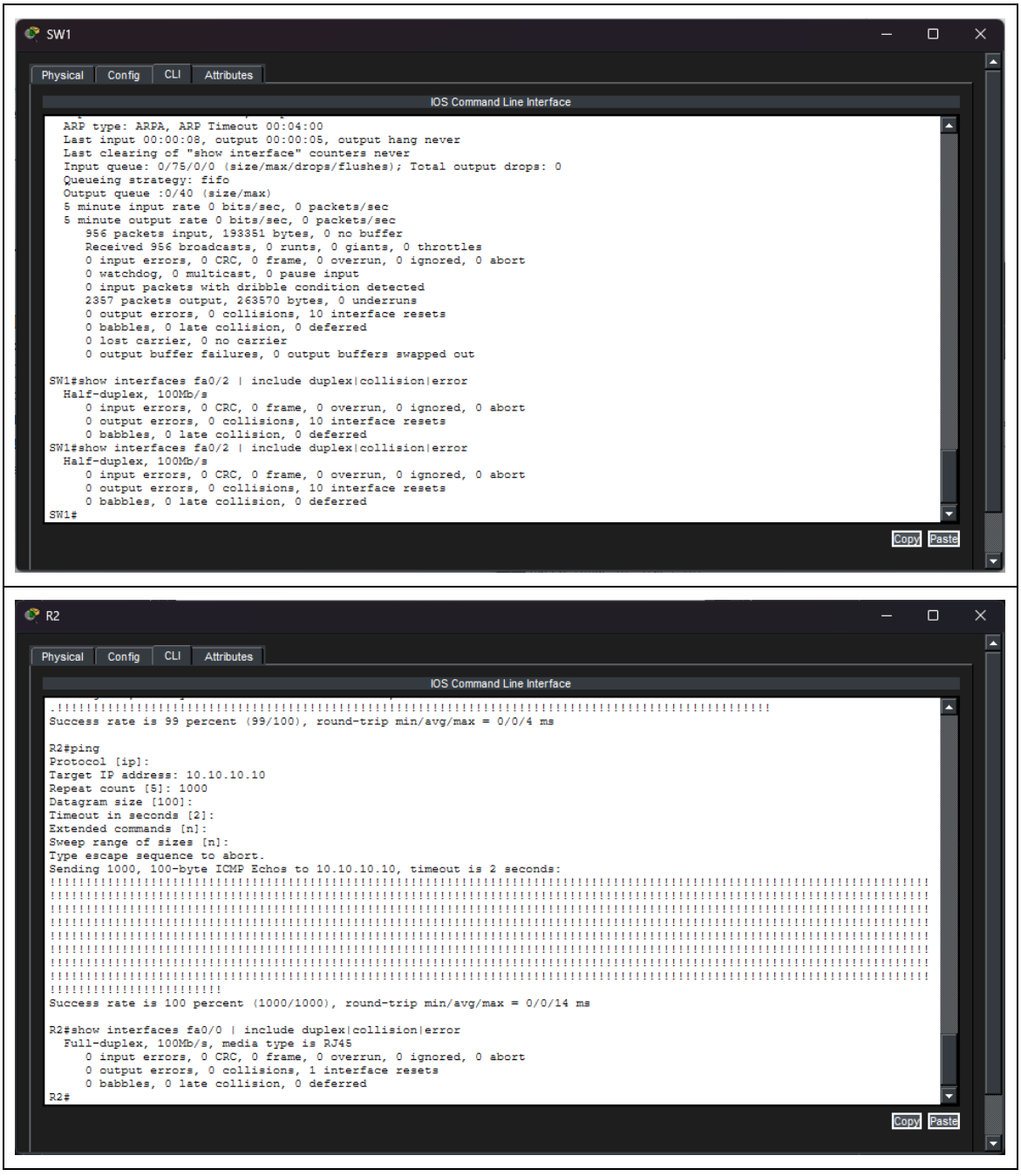
Task 41 — Shutdown Interface



Task 42 — Restore Interface



Task 43 — Half-Duplex



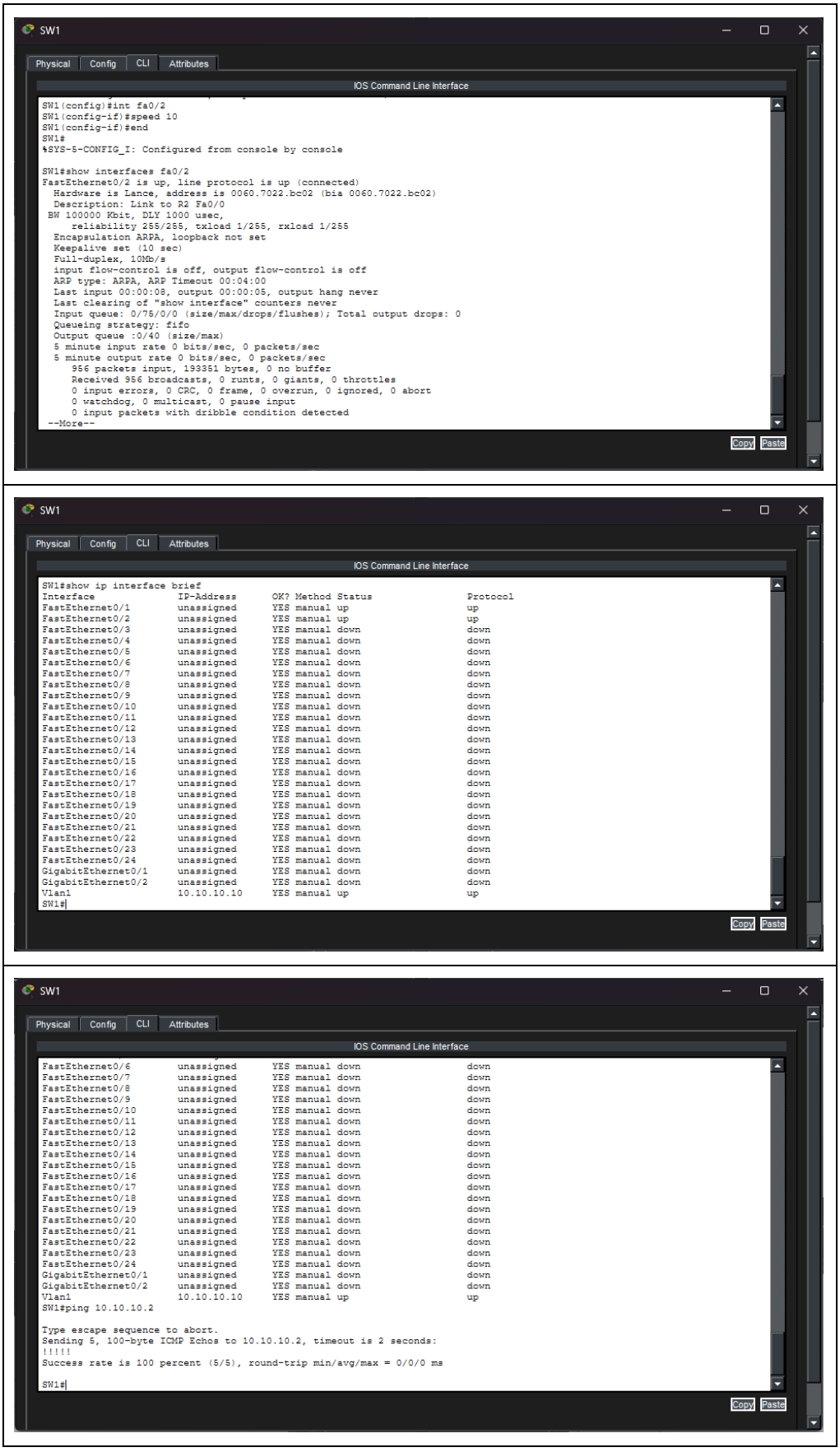
R2
Physical
Config
CLI
Attributes
IOS Command Line Interface
.!!!

Task 44 — Full-Duplex



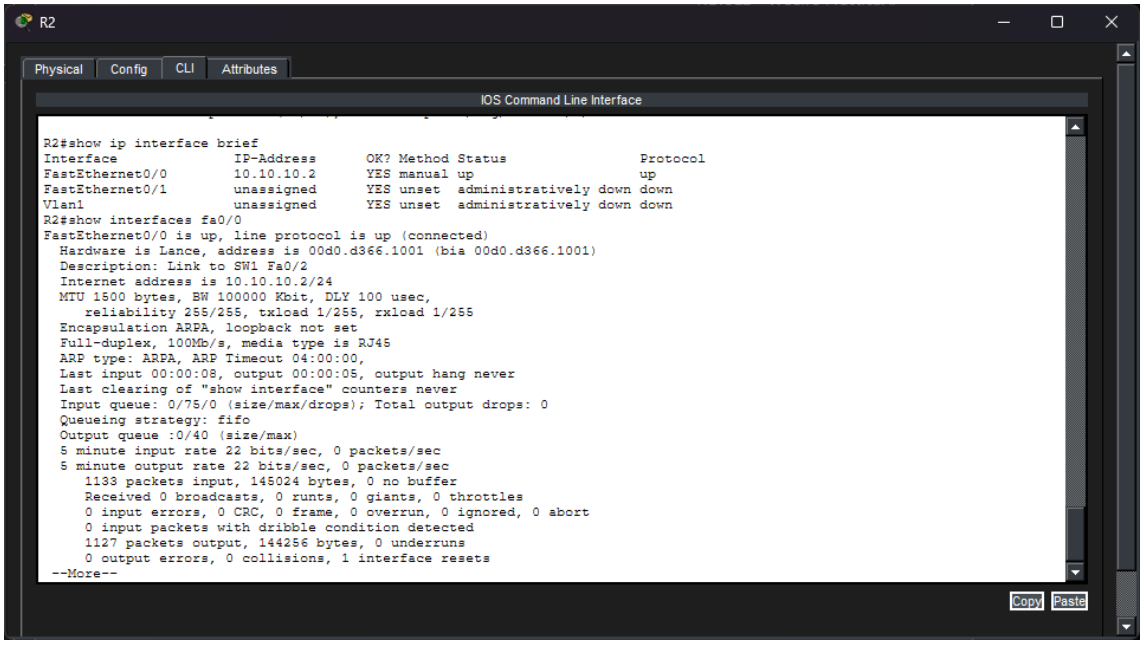
R2
Physical
Config
CLI
Attributes
IOS Command Line Interface
!!!

Task 45 — Set Speed



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20261193

Task 46 — Interface Status



Reflection Questions

Task 47 – Duplex Mismatch

In my instance with the duplex mismatch no changes were seen between SW1 and R2. The interface status/brief changes to show half-duplex or the speed change, but traffic remained unaffected, even when R2 sent 1000 repeated pings to 10.10.10.10, there were no errors, no fails or collisions.